



Systems, Control and Robotics

Courses for Systems, Control and Robotics

The two-year master's programme in Systems, Control and Robotics consists of three terms of courses and one final term dedicated to the master's degree project. Each term consist of approximately 30 ECTS credits. Depending on which track you choose, you will study different courses. The courses presented on this page apply to studies starting in autumn 2026.

Year 1

The following course requirements need to be satisfied to get a degree:

- all common mandatory courses of 30 credits
- all mandatory courses of 15 credits depending on track
- at least 21 credits of the conditionally elective courses depending on the track
- at least one project course completed
- degree project equivalent to 30 credits completed
- freely elective courses up to 120 credits

Compulsory and conditionally elective courses from one track are recommended on all other tracks.

A course can only be listed once below and is listed only as compulsory if it is compulsory on one track and only conditionally elective on another. Please use the course list for each separate track for a complete account of the conditionally elective courses.

Mandatory courses for all tracks

- [Introduction to Robotics \(DD2410\) 7.5 credits](#)
- [The Sustainable Systems and Control Engineer \(EL2220\) 3 credits](#)
- [Modelling of Dynamical Systems \(EL2820\) 7.5 credits](#)
- [Control Theory and Practice, Advanced Course \(EL2520\) 7.5 credits](#)

Recommended courses for all track

- [Program System Construction Using C++ \(DD1388\) 7.5 credits](#)
- [Project Course in Robotics and Autonomous Systems \(DD2419\) 9 credits](#)
- [Probabilistic Graphical Models \(DD2420\) 7.5 credits](#)
- [Deep Learning in Data Science \(DD2424\) 7.5 credits](#)
- [Artificial Intelligence and Multi Agent Systems \(DD2438\) 15 credits](#)
- [Electricity Market Analysis \(EG2210\) 7.5 credits](#)
- [Building Networked Systems Security \(EP2520\) 7.5 credits](#)
- [Speech and Audio Processing \(EQ2321\) 7.5 credits](#)
- [Embedded Software \(IL2212\) 7.5 credits](#)
- [Technical Communication in English \(LS2444\) 7.5 credits](#)
- [Applied Vehicle Dynamics Control \(SD2231\) 7.5 credits](#)
- [Applied Linear Optimization \(SF2812\) 7.5 credits](#)
- [Geometric Control Theory \(SF2842\) 7.5 credits](#)
- [Probability Theory \(SF2940\) 7.5 credits](#)
- [Time Series Analysis \(SF2943\) 7.5 credits](#)
- [Optimization \(SF1861\) 6 credits](#)
- [Hybrid and Embedded Control Systems \(EL2450\) 7.5 credits](#)
- [Project Course in Data Science \(DD2430\) 7.5 credits](#)
- [Multimodal Interaction and Interfaces \(DT2140\) 7.5 credits](#)
- [Project in Conversational Systems \(DT2151\) 7.5 credits](#)
- [Business Development and Quality Management \(EH2030\) 7.5 credits](#)
- [Management of Projects \(EH2720\) 7.5 credits](#)
- [Build your own Radar System, Project Course \(EK2370\) 7.5 credits](#)
- [Automatic Control, General Course \(EL1010\) 6 credits](#)
- [Automatic Control, Project Course, Smaller Course \(EL2425\) 7.5 credits](#)
- [Nonlinear Control \(EL2620\) 7.5 credits](#)
- [Model Predictive Control \(EL2700\) 7.5 credits](#)
- [Reinforcement Learning \(EL2805\) 7.5 credits](#)
- [Signal Theory \(EQ1220\) 7.5 credits](#)
- [Digital Communications \(EQ2310\) 9 credits](#)
- [Embedded Systems \(IL2206\) 7.5 credits](#)
- [Essential STEM communication in English \(LS1452\) 7.5 credits](#)
- [Rhetoric - the Art of Persuasion \(LS1464\) 7.5 credits](#)
- [English for Employment \(LS2442\) 7.5 credits](#)
- [Industrial Management, Basic Course \(ME1003\) 6 credits](#)
- [Dynamics and Motion Control \(MF2007\) 9 credits](#)
- [Mechatronics basic Course \(MF2030\) 6 credits](#)
- [Robust Mechatronics \(MF2043\) 6 credits](#)

- [Optimization \(SF1811\) 6 credits](#)
- [Mathematical Systems Theory \(SF2832\) 7.5 credits](#)
- [Complex Analysis \(SF1691\) 7.5 credits](#)
- [Project in Cognitive Systems \(DT2150\) 7.5 credits](#)

Compulsory and conditionally elective courses from one track are recommended on all other tracks.

A course can only be listed once below and is listed only as compulsory if it is compulsory on one track and only conditionally elective on another. Please use the course list for each separate track for a complete account of the conditionally elective courses.

Choose at least 21 credits from the list of conditionally elective courses.

Mandatory courses

- [Hybrid and Embedded Control Systems \(EL2450\) 7.5 credits](#)

Conditionally elective courses

- [Machine Learning \(DD2421\) 7.5 credits](#)
- [Deep Learning in Data Science \(DD2424\) 7.5 credits](#)
- [Artificial Neural Networks and Deep Architectures \(DD2437\) 7.5 credits](#)
- [Machine Learning Theory \(EL2810\) 7.5 credits](#)
- [Adaptive Signal Processing \(EQ2401\) 7.5 credits](#)
- [Complex Analysis \(SF1691\) 7.5 credits](#)
- [Applied Nonlinear Optimization \(SF2822\) 7.5 credits](#)
- [Geometric Control Theory \(SF2842\) 7.5 credits](#)
- [Artificial Intelligence \(DD2380\) 6 credits](#)
- [Safe Robot Planning and Control \(DD2416\) 7.5 credits](#)
- [Machine Learning, Advanced Course \(DD2434\) 7.5 credits](#)
- [Robot Learning and Embodied AI \(DD2600\) 7.5 credits](#)
- [Automatic Control, Project Course, Smaller Course \(EL2425\) 7.5 credits](#)
- [Nonlinear Control \(EL2620\) 7.5 credits](#)
- [Reinforcement Learning \(EL2805\) 7.5 credits](#)
- [Cyber-Physical Security in Time-Critical Systems \(EL2850\) 7.5 credits](#)
- [Digital Signal Processing \(EQ2300\) 7.5 credits](#)

- [Mathematical Systems Theory \(SF2832\) 7.5 credits](#)

Compulsory and conditionally elective courses from one track are recommended on all other tracks.

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Choose at least 21 credits from the list of conditionally elective courses.

Mandatory courses

- [Image Analysis and Computer Vision \(DD2423\) 7.5 credits](#)
- [Applied Estimation \(EL2320\) 7.5 credits](#)

Conditionally elective courses

- [Deep Learning, advanced course \(DD2610\) 7.5 credits](#)
- [Speech and Audio Processing \(EQ2321\) 7.5 credits](#)
- [Research project in Robotics, Perception and Learning \(DD2411\) 15 credits](#)
- [Engineering project in Robotics, Perception and Learning \(DD2414\) 15 credits](#)
- [Project Course in Robotics and Autonomous Systems \(DD2419\) 9 credits](#)
- [Machine Learning \(DD2421\) 7.5 credits](#)
- [Deep Learning in Data Science \(DD2424\) 7.5 credits](#)
- [Artificial Neural Networks and Deep Architectures \(DD2437\) 7.5 credits](#)
- [Artificial Intelligence and Multi Agent Systems \(DD2438\) 15 credits](#)
- [Speech and Speaker Recognition \(DT2119\) 7.5 credits](#)
- [Machine Learning Theory \(EL2810\) 7.5 credits](#)
- [Dynamics and Motion Control \(MF2007\) 9 credits](#)
- [Hybrid and Embedded Control Systems \(EL2450\) 7.5 credits](#)
- [Artificial Intelligence \(DD2380\) 6 credits](#)
- [Social Robotics \(DD2413\) 7.5 credits](#)
- [Safe Robot Planning and Control \(DD2416\) 7.5 credits](#)

- [Machine Learning, Advanced Course \(DD2434\) 7.5 credits](#)
- [Robot Learning and Embodied AI \(DD2600\) 7.5 credits](#)
- [Automatic Control, Project Course, Smaller Course \(EL2425\) 7.5 credits](#)
- [Nonlinear Control \(EL2620\) 7.5 credits](#)
- [Model Predictive Control \(EL2700\) 7.5 credits](#)
- [Reinforcement Learning \(EL2805\) 7.5 credits](#)
- [Cyber-Physical Security in Time-Critical Systems \(EL2850\) 7.5 credits](#)
- [Digital Signal Processing \(EQ2300\) 7.5 credits](#)
- [Applied Vehicle Dynamics Control \(SD2231\) 7.5 credits](#)

Year 2

The following course requirements need to be satisfied to get a degree:

- all common mandatory courses of 30 credits
- all mandatory courses of 15 credits depending on track
- at least 21 credits of the conditionally elective courses depending on the track
- at least one project course completed
- degree project equivalent to 30 credits completed
- freely elective courses up to 120 credits

Compulsory and conditionally elective courses from one track are recommended on all other tracks.

A course can only be listed once below and is listed only as compulsory if it is compulsory on one track and only conditionally elective on another. Please use the course list for each separate track for a complete account of the conditionally elective courses.

Mandatory courses for all tracks

- [The Sustainable Systems and Control Engineer \(EL2220\) 3 credits](#)
- [Degree Project in Computer Science and Engineering, specializing in Systems, Control and Robotics, Second Cycle \(DA236X\) 30 credits](#)
- [Degree Project in Electrical Engineering, specializing in Systems, Control and Robotics, Second Cycle \(EA236X\) 30 credits](#)
- [Theory and Methodology of Science \(Natural and Technological Science\)](#)

[\(AK2030\) 4.5 credits](#)

Recommended courses for all tracks

- [Algorithms and Complexity \(DD2352\) 7.5 credits](#)
- [Neuroscience \(DD2401\) 7.5 credits](#)
- [Project Course in Robotics and Autonomous Systems \(DD2419\) 9 credits](#)
- [Probabilistic Graphical Models \(DD2420\) 7.5 credits](#)
- [Mathematical Modelling of Biological Systems \(DD2435\) 9 credits](#)
- [Artificial Intelligence and Multi Agent Systems \(DD2438\) 15 credits](#)
- [Statistical Methods in Applied Computer Science \(DD2447\) 6 credits](#)
- [Software Reliability \(DD2459\) 7.5 credits](#)
- [Bigger Advanced, Individual Course in Computer Science \(DD2464\) 9 credits](#)
- [Multimodal Interaction and Interfaces \(DT2140\) 7.5 credits](#)
- [Computer Applications and Machine Learning in Electric Power Systems \(EG2140\) 7.5 credits](#)
- [Business Development and Quality Management \(EH2030\) 7.5 credits](#)
- [Management of Projects \(EH2720\) 7.5 credits](#)
- [Build your own Radar System, Project Course \(EK2370\) 7.5 credits](#)
- [Automatic Control, Project Course, Smaller Course \(EL2425\) 7.5 credits](#)
- [Hybrid and Embedded Control Systems \(EL2450\) 7.5 credits](#)
- [Nonlinear Control \(EL2620\) 7.5 credits](#)
- [Model Predictive Control \(EL2700\) 7.5 credits](#)
- [Reinforcement Learning \(EL2805\) 7.5 credits](#)
- [Building Networked Systems Security \(EP2520\) 7.5 credits](#)
- [Digital Communications \(EQ2310\) 9 credits](#)
- [Speech and Audio Processing \(EQ2321\) 7.5 credits](#)
- [Adaptive Signal Processing \(EQ2401\) 7.5 credits](#)
- [Cyber-Physical Networking \(EQ2871\) 7.5 credits](#)
- [Embedded Systems \(IL2206\) 7.5 credits](#)
- [Embedded Software \(IL2212\) 7.5 credits](#)
- [German B2 for Engineers \(LS2426\) 7.5 credits](#)
- [French B2 for Engineers \(LS2436\) 7.5 credits](#)
- [Spanish B2 for Engineers \(LS2449\) 7.5 credits](#)
- [Leadership in Cross-Cultural and Industrial Contexts \(ME2089\) 6 credits](#)
- [Dynamics and Motion Control \(MF2007\) 9 credits](#)
- [Mechatronics basic Course \(MF2030\) 6 credits](#)
- [Robust Mechatronics \(MF2043\) 6 credits](#)
- [Applied Vehicle Dynamics Control \(SD2231\) 7.5 credits](#)
- [Parallel Computations for Large- Scale Problems \(SF2568\) 7.5 credits](#)

- [Applied Linear Optimization \(SF2812\) 7.5 credits](#)
- [Mathematical Systems Theory \(SF2832\) 7.5 credits](#)
- [Geometric Control Theory \(SF2842\) 7.5 credits](#)
- [Optimal Control Theory \(SF2852\) 7.5 credits](#)
- [Probability Theory \(SF2940\) 7.5 credits](#)
- [Time Series Analysis \(SF2943\) 7.5 credits](#)
- [Software Engineering \(DD1385\) 6 credits](#)
- [Rhetoric - the Art of Persuasion \(LS1464\) 7.5 credits](#)
- [Industrial Management, Basic Course \(ME1003\) 6 credits](#)
- [Complex Analysis \(SF1691\) 7.5 credits](#)
- [Optimization \(SF1811\) 6 credits](#)
- [Optimization \(SF1861\) 6 credits](#)
- [Geometric Robot Learning \(DD2603\) 7.5 credits](#)

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Choose at least 21 credits from the list of conditionally elective courses.

Mandatory courses

- [Model Predictive Control \(EL2700\) 7.5 credits](#)

Conditionally elective courses

- [Artificial Intelligence \(DD2380\) 6 credits](#)
- [Safe Robot Planning and Control \(DD2416\) 7.5 credits](#)
- [Machine Learning \(DD2421\) 7.5 credits](#)
- [Deep Learning in Data Science \(DD2424\) 7.5 credits](#)
- [Machine Learning, Advanced Course \(DD2434\) 7.5 credits](#)
- [Artificial Neural Networks and Deep Architectures \(DD2437\) 7.5 credits](#)
- [Robot Learning and Embodied AI \(DD2600\) 7.5 credits](#)
- [Deep Generative Models and Synthesis \(DD2601\) 7.5 credits](#)
- [Deep Learning, advanced course \(DD2610\) 7.5 credits](#)
- [Automatic Control, Project Course, Smaller Course \(EL2425\) 7.5 credits](#)
- [Nonlinear Control \(EL2620\) 7.5 credits](#)
- [Reinforcement Learning \(EL2805\) 7.5 credits](#)
- [Machine Learning Theory \(EL2810\) 7.5 credits](#)
- [Cyber-Physical Security in Time-Critical Systems \(EL2850\) 7.5 credits](#)
- [Digital Signal Processing \(EQ2300\) 7.5 credits](#)
- [Adaptive Signal Processing \(EQ2401\) 7.5 credits](#)
- [Optimal Filtering \(EQ2801\) 7.5 credits](#)

- [Applied Nonlinear Optimization \(SF2822\) 7.5 credits](#)
- [Mathematical Systems Theory \(SF2832\) 7.5 credits](#)
- [Geometric Control Theory \(SF2842\) 7.5 credits](#)
- [Optimal Control Theory \(SF2852\) 7.5 credits](#)
- [Complex Analysis \(SF1691\) 7.5 credits](#)

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Choose at least 21 credits from the list of conditionally elective courses.

Conditionally elective courses

- [Research project in Robotics, Perception and Learning \(DD2411\) 15 credits](#)
- [Artificial Intelligence \(DD2380\) 6 credits](#)
- [Social Robotics \(DD2413\) 7.5 credits](#)
- [Safe Robot Planning and Control \(DD2416\) 7.5 credits](#)
- [Project Course in Robotics and Autonomous Systems \(DD2419\) 9 credits](#)
- [Machine Learning \(DD2421\) 7.5 credits](#)
- [Machine Learning, Advanced Course \(DD2434\) 7.5 credits](#)
- [Artificial Neural Networks and Deep Architectures \(DD2437\) 7.5 credits](#)
- [Robot Learning and Embodied AI \(DD2600\) 7.5 credits](#)
- [Deep Generative Models and Synthesis \(DD2601\) 7.5 credits](#)
- [Deep Learning, advanced course \(DD2610\) 7.5 credits](#)
- [Automatic Control, Project Course, Smaller Course \(EL2425\) 7.5 credits](#)
- [Hybrid and Embedded Control Systems \(EL2450\) 7.5 credits](#)
- [Nonlinear Control \(EL2620\) 7.5 credits](#)
- [Model Predictive Control \(EL2700\) 7.5 credits](#)
- [Reinforcement Learning \(EL2805\) 7.5 credits](#)
- [Machine Learning Theory \(EL2810\) 7.5 credits](#)
- [Cyber-Physical Security in Time-Critical Systems \(EL2850\) 7.5 credits](#)
- [Digital Signal Processing \(EQ2300\) 7.5 credits](#)

- [Speech and Audio Processing \(EQ2321\) 7.5 credits](#)
- [Dynamics and Motion Control \(MF2007\) 9 credits](#)

Systems, Control and Robotics

[Introduction](#)

[Courses](#)

[Entry requirements](#)

[Fees and scholarships](#)

[Contact](#)

Admissions

[How to apply](#)

[Entry requirements](#)

[Fees](#)

[Scholarships](#)

Contact

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[Career after KTH](#)

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[Cost of living](#)

[Accommodation](#)

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Changes in the programme may occur.